



Science SQUAD



A Multi-Skilled Book on Science Education

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(M. Tech)

1



I am.



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FORWARD

This book **Science Squad** is prepared in accordance with the Latest Syllabus prescribed by CBSE, following the needs of students' studies. The concepts are described in a well-structured and systematic manner.

As per the **National Education Policy (NEP)** and **National Curriculum Framework (NCF)**, there is a need to switch from old learning to Competency-Based Learning. This will help the students develop critical thinking and problem-solving skills.

The beginning of the chapter has a heading named **Learning Outcomes**. This gives an idea of the content of the chapter. **Facts** offer different types of unknown facts. **Backpedal** gives an overall summary of the chapter. It explains the chapter in a nutshell. Then comes the **Testing Time**. It includes the exercises based on the text studied and discussed.

Think and Respond, Life Skills and Values, Art Integration (Activity), Understanding Skills, **Funny Projects** are also given.

We sincerely hope that our book is an effective product in the hands of students for better understanding and knowledge of the subject preparing them for their examinations.

We welcome your valuable feedback to improve the quality of our books.

— Author and Publishers



The **National Education Policy** creates a new dimension of joyful learning along with completely mapped parameters of education in the new era.





CHAPTER SKETCH

Learning Outcomes includes the topics to be learned in the chapter.



LEARNING OUTCOMES

- ▶ Kinds of Plants
- ▶ Parts of a Plant
- ▶ The Roots
- ▶ The Stem
- ▶ The Leaf
- ▶ The Flowers
- ▶ Fruits and Seeds

Facts includes some facts related to the chapter to enhance the knowledge of the students.



There are some organisms which decomposes the dead bodies of animals and plants left by scavengers. They are called decomposers.

Click Your Brain provides a short exercise for gaining a strong command on the topic.

Click Your Brain

Name the organs involved in the given organ system.

1. Skeletal system _____
2. Digestive system _____
3. Muscular system _____
4. Respiratory system _____

Keys provides the extended meanings of important words from the chapter to understand the topic.

KEY WORDS

- | | | |
|--------|---|------------------------------|
| Decay | — | to rot away |
| Gravel | — | mixture of rocks and pebbles |

Backpedal contains important summary points to recall the chapter at a glance before doing exercises.

Backpedal

- ▶ We have external and internal organs.
- ▶ We have five sense organs — eyes, ears, nose, tongue and skin.
- ▶ Organs come together to make up an organ system.
- ▶ We have the following organ systems:
 - (i) The skeletal system (ii) The muscular system (iii) The digestive system
 - (iv) The respiratory system (v) The nervous system (vi) The circulatory system
 - (vii) The excretory system (viii) The reproductive system

Testing Time contains exercises to test knowledge on the chapter.

TESTING TIME

Objective Type Questions

Continuous Review

A. Tick (✓) the correct option:

- | | | | |
|--|--------------|---------------------|---------------------|
| 1. Matter _____ | (i) has mass | (ii) occupies space | (iii) Both of these |
| 2. Matter has _____ states. | (i) one | (ii) two | (iii) three |
| 3. Solids have definite / fixed _____. | (i) shape | (ii) volume | (iii) Both of these |
| 4. Gases can _____. | (i) flow | (ii) fly | (iii) None of these |

Think and Respond to making answers to some hard and practical questions.

Think and Respond

- An elephant has long teeth outside its mouth? How elephants eat their food?

Critical Thinking

Life Skills and Values related to life skills and values questions.

Life skills and Values

Environmental Literacy

- We are cutting trees in large numbers. It is causing deforestation. What can you do to check it?



Funny Projects provides interesting experiments based on the chapter content.

FUNNY PROJECTS

- Explain the uses of water through a flowchart.
- Draw the picture showing the water cycle in your notebook.

Problem Solving
Art Integration



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1

LIVING AND NON-LIVING THINGS



LEARNING OUTCOMES

- ▶ Living things and non-living things
- ▶ Natural things and man-made things

There are many different things around us : humans, animals, plants, stones, water, and air. All these things around us are mainly divided into two groups.

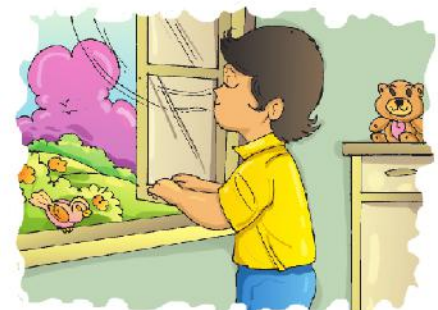
LIVING THINGS AND NON-LIVING THINGS

Things that breathe, feel, and move on their own are called **living things**.

Living things take food and water, grow, reproduce, become old and die. All humans, animals, birds, and plants are living things. They have life.

We are living things. We have life.

1. We **breathe** with our nose.
2. We feel pain when we get hurt.
3. If it is cold, we can feel it.



We can **feel** the cold through our skin.



When we feel cold, we wear a woollen sweater.

Things that do not breathe, feel, or move on their own are called **non-living things**.

Non-living things like a teddy do not breathe or feel cold.

4. Living things can **move** from one place to another on their own.

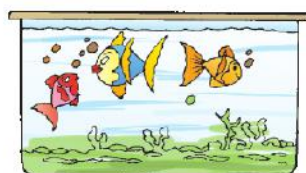
They may walk, run, swim, or fly.



Walking



Running



Swimming

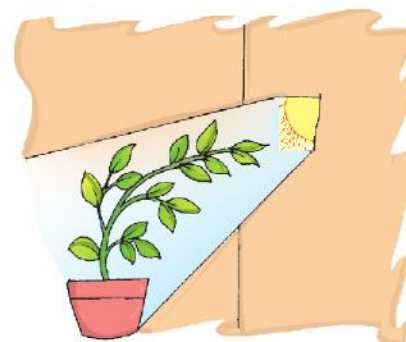
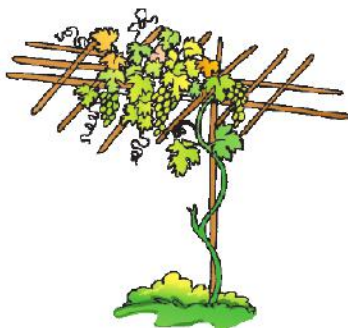
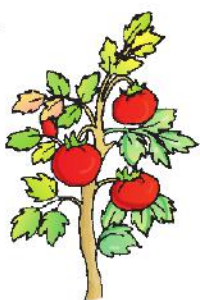


Flying

Plants move upwards while growing for more sunlight.

The grapevine climbs along a support for more sunlight.

A plant in a room turns towards light.



Table, chairs, pens, fans, bricks, and stone are non-living things.

They move only when someone moves them.

Non-living things do not move on their own.

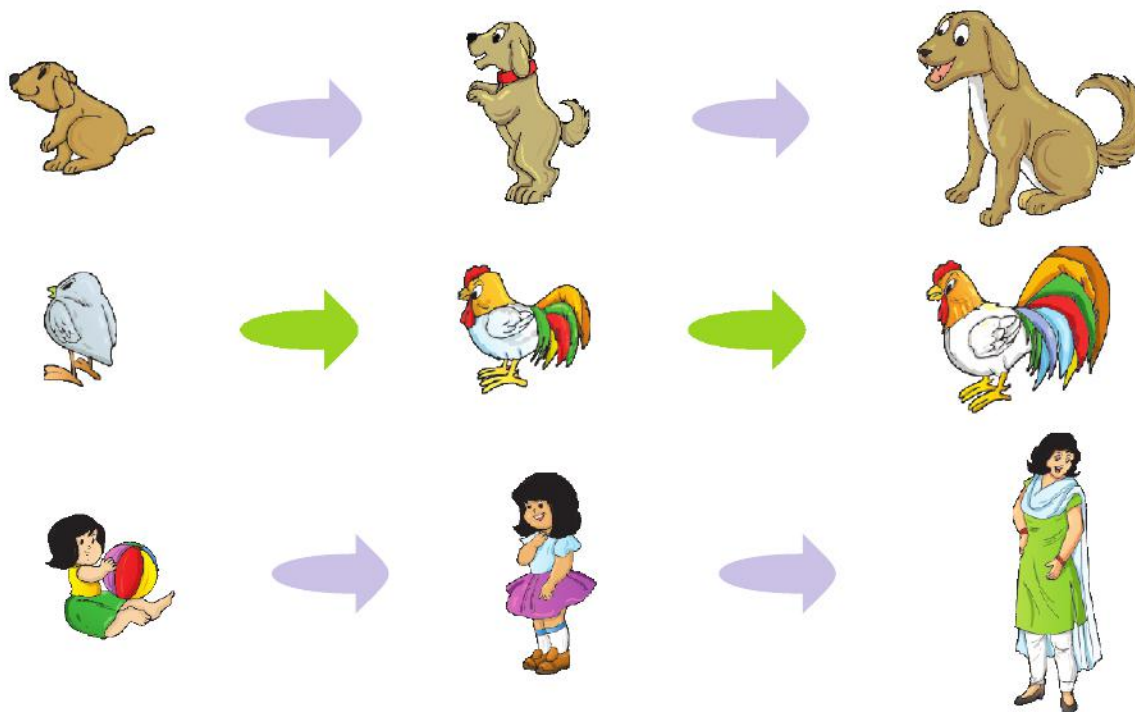


People and animals eat food for energy. They need food to grow, move and do many other activities.

5. Living things **need and take food and water to grow**.

A small pup takes food and water. It grows to become a dog.

6. A chick grows into a hen. A baby grows into a child.



A small seed takes food and water from the soil.

It grows into a sapling, which grows into a bigger plant.

Plants do not move from place to place.



Non-living things do not need to air, food, and water.

They do not grow, move, feel, or have babies.

7. Living things **reproduce** young ones who look like them.

A dog reproduces small dogs.



A wheat plant produces seeds for new plants.

Non-living things do not reproduce.

8. All living things **die** when they become old.

A dog becomes old and dies.

A plant becomes old and dies.



When a living thing dies, it can not breathe, feel, or move. It does not take food and water, grow, or reproduce. It has no life.



Plants breathe through tiny pores on their leaves called **stomata**.

NATURAL AND MAN-MADE THINGS

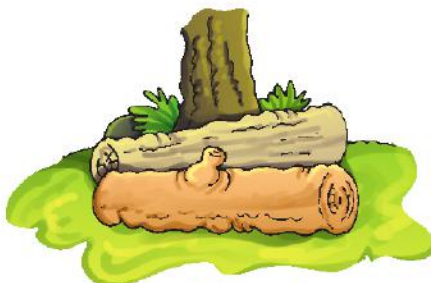
Non-living things can be natural or man-made things sun, clouds, water, and stones are found in nature.

Non-living things, not made by man, are called **natural things**.

Rocks, wood, and water are found in nature. They are not made by man. They are natural things.



Rocks



Wood



Water

Non-living things made by man are called **man-made things**.

House, cars, clothes, pens, paper are made by man.

They are not found in nature. They are man-made things.



House



Car



Clothes



Pen-paper

Click Your Brain

1. What are living things?

2. What are natural things?

3. What are man-made things?

KEY WORDS

- | | |
|-------------------|---|
| Living Things | — Things that breathe, feel, move, eat, grow, reproduce, and die. |
| Non-living Things | — Which do not breathe, feel, or move. |
| Man-made Things | — Non-living things made by man. |
| Reproduce | — To give birth to young ones or to lay eggs. |

Backpedal

- ▶ Things around us are either living or non-living.
- ▶ Living things breathe, feel, move, take food, grow and reproduce.
- ▶ All humans, animals, and plants are living things.

- ▶ Non-living things do not breathe, feel, move, take food, grow or reproduce.
- ▶ Natural things are found in nature and are not made by man.



Continuous Review

Objective Type Questions

A. Tick (✓) the correct option:

- Which of these is not a non-living thing?
 (i) Plant ☐ (ii) Pencil ☐ (iii) rabbit ☐
- Living things :
 (i) move ☐ (ii) feel cold ☐ (iii) both (i) and (ii) ☐
- Which of these is not a living thing?
 (i) tiger ☐ (ii) stone ☐ (iii) rose plant ☐
- A plant will grow into a :
 (i) baby ☐ (ii) tree ☐ (iii) cat ☐

B. Fill in the blanks with correct words:

- A seed grows into a _____.
- We see so many _____ around us.
- _____ is a natural non-living thing.

Things
Water
Plant

C. Write True or False:

- All living things do not need food.
- A plant grows, so it is a living thing.
- A pen requires food.
- A car moves, so it is a living thing.

Subjective Type Questions :

A. Answer in one word:

- A thing that does not grow.

2. A car is a living/non-living thing.
3. A cat/teddy takes food.
4. Water is a natural/man-made thing.

B. Answer the following questions:

1. Define living and non-living things.
2. What are the kinds of non-living things? Name them.
3. Compare living and non-living things.

Think and Respond

- Do you feed your teddy like your puppy? Why? **Critical Thinking**

Life skills and Values



Social Literacy

- Mother said to the child, "You will become as big as I one day." Was she right? Why?

FUNNY PROJECTS



- Answer the questions about each thing. Write in the table below.

Experiential Learning

	Does it move on its own?	What does it eat?	Does it grow?	Is it a living thing?
Parrot				
Ant				
Plant				
Teddy				



2

PLANT KINGDOM



LEARNING OUTCOMES

- ▶ Plants
- ▶ Care of Plants
- ▶ Parts of a Plant
- ▶ Importance of Plants
- ▶ Types of Plants

PLANTS

You see greenery in the fields, hills and parks. It is due to the plants. Plants are big or small. They are living things. They make their food on their own. They give us flowers, fruits and vegetables, etc. Plants have different parts. Let's learn about them.



There are some plants that are not green.

PARTS OF A PLANT

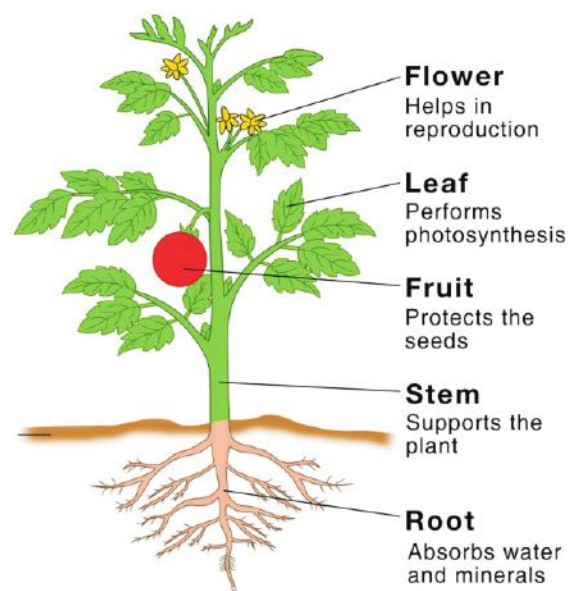
Roots : **Roots** grow under the soil. They keep the plant firm and fixed above the ground. They collect water and minerals from the soil.

Stem : **Stem** bears branches, leaves, flowers and fruits. It carries water and minerals to all parts of the plant. It is the hardest part of the plant.

Branches : **Branches** grow on the stem in all directions. They bear leaves, flowers and fruits.

Leaves : **Leaves** are the green part of a plant. They make food for the plant. They are called **the kitchen of the plant**. Some plants have big leaves and some have small leaves.

Flowers : **Flowers** are the bright and colourful part of a plant. They make fruits and seeds.



Parts of a Plant



Rafflesia Arnoldii is the world's largest flower.

Fruits : **Fruits** are the edible parts of the plant. But all the fruits are not edible. They have seeds in them. Some fruits have one seed like mango and some have many seeds like apple.



TYPES OF PLANTS

Plants are categorised on the basis of their sizes and shapes. They are trees, shrubs, herbs, climbers and creepers.

Trees : Trees are tall, big and strong plants. They have thick and strong stem. Neem, peepal, mango, banyan, coconut are trees.



Mango



Peepal



Neem



Banyan



Coconut

Shrubs : Shrubs are small plants. They are not as strong as trees. Their stems are hard and woody. Rose, cotton, lemon, tulsi, henna, etc. are shrubs.



Rose



Cotton



Lemon



Tulsi



Henna

Herbs : **Herbs** are smaller than shrubs. They have weak and soft stems. They grow just above the ground. Coriander, wheat, tomato, mustard and mint are herbs.



Coriander



Tomato



Mint



Mustard



Wheat

Climbers : Climbers are weak plants. They have green and tender stems. They need support to stand straight or climb. Peas, grapevine, money plant, cucumber, gourd, etc. are **climbers**.



Peas



Grapevine



Money plant



Cucumber



Gourd

Creepers : Creepers are very weak plants. They grow along the ground. Watermelon, pumpkin, muskmelon, caltrop, etc. are **creepers**.



Pumpkin



Muskmelon



Watermelon



Caltrop

Other Plants : There are some other plants also. They grow in water. They are called **water** or aquatic plants. They are lotus, water lily, water hyacinth, hydrilla, pistia, etc. Some plants like cactus grow in sand, on rocks and in hot places with little water. Tulip and fir trees grow in cold places and on the mountains.



Lotus



Water Lily



Water Hyacinth



Hydrilla



Pistia

Thorns : There are some thorny plants. They are rose, lemon and bush. Thorns protect the plants from animals.

CARE OF PLANTS

We should take care of plants. We should not damage the branches of plants. We should not pluck the flowers and leaves. We should avoid walking on the grass when we visit the parks.

Click Your Brain

Name these.

- | | | | |
|-----------------|-------|-------|-------|
| 1. Trees | _____ | _____ | _____ |
| 2. Shrubs | _____ | _____ | _____ |
| 3. Herbs | _____ | _____ | _____ |
| 4. Climbers | _____ | _____ | _____ |
| 5. Creepers | _____ | _____ | _____ |
| 6. Water Plants | _____ | _____ | _____ |

IMPORTANCE OF PLANTS

1. Plants make the environment green and clean.
2. They give us fruits and vegetables.

3. They give us oxygen to breathe.
4. They give us wood for different purposes.

KEY WORDS

Roots	—	part of a plant that grows under the soil
Trees	—	tall, big and strong plants
Shrubs	—	small plants with hard, woody stem
Herbs	—	smaller plants than shrubs
Creepers	—	plants that grow along the ground
Climbers	—	plants that need support to stand straight



Backpedal

- ▶ Plants are green living things.
- ▶ Plants are of different kinds, like trees, shrubs, herbs, climbers and creepers.
- ▶ There are some plants which grow in water.
- ▶ Plants like cactus grow in sandy and hot places.
- ▶ Thorns protect the plants from animals.
- ▶ Leaves are called the kitchen of the plant.



Objective Type Questions

Continuous Review

A. Tick (✓) the correct option:

1. Plants are _____.

(i) green



(ii) yellow



(iii) grapevine



2. This grows under the soil.

(i) Stem

(ii) Flower

(iii) Root

3. This is a climber.

(i) Neem

(ii) Rose

(iii) Pumpkin

B. Fill in the blanks with correct words:

1. Plants make their _____ on their own.

2. Roots keep the plant _____ and fixed.

3. Leaves are the _____ of a plant.

4. Plants are very _____ for us.

kitchen
important
firm
food

C. Write True or False:

1. Roots collect water and minerals from the soil.

2. Branches bear leaves, flowers and fruits.

3. Stem is the weak part of a plant.

4. Plants are not important for us.

Subjective Type Questions

A. Answer in one word:

1. What grows under the soil?

2. Where do fruits and flowers grow?

3. What is called the kitchen of a plant?

4. What does a plant give us?

B. Answer the following questions:

1. What are plants?

2. Define root and stem.

3. What are the parts of a plant?

4. How many types of plants are there?

5. What are water plants?

6. How are plants important for us?

Think and Respond

- You know that plants and trees give us food. If there were no plants, what would we eat?

Critical Thinking

Life skills and Values



Environmental Literacy

- Plants are very important for us. But we are cutting them excessively. How can we protect them?

FUNNY PROJECTS



- Paste the pictures of some climbers in your scrapbook and write their names.
- Name the given trees.

Curiosity

Recognising Skills









- Make a chart of different kinds of plants and hang it in your classroom.

Art Integration



3

HOW PLANTS GROW?



LEARNING OUTCOMES

- ▶ Flowers
- ▶ Seeds
- ▶ Growing of a Seed into a Plant

Plants are very important part of our lives. They give us many things. But how do they grow?

Plants grow from seeds. Seeds are found in fruits.



Not all flowers have nice fragrance.

FLOWERS

Most of the plants bear flowers. Flowers are of different kinds. They are of many colours and shapes. They have sweet smell, so they are used in perfumes, soaps and cosmetics. Some flowers are rose, jasmine, sunflower, tulip, etc. These flowers do not become fruits.



Rose



Jasmine



Tulip



Sunflower

Some flowers become fruits. They are mango, guava, neem, chilly, tomato, etc.



Mango



Guava



Neem



Tomato

When the flower fades away or gets dried, fruit starts growing. It has a seed or many seeds. These seeds grow into new plants.



SEEDS

Almost all the fruits have seeds. Some fruits have one seed only like mango, while some fruits have many seeds like watermelon.



Click Your Brain

Mango has one seed. Name some fruits with one seed.

Watermelon has many seeds. Name some fruits with many seeds.

We do not eat seeds of some fruits. We throw away these seeds. But we eat fruits with their seeds, like guava, kiwi, strawberry, corn, etc.



Guava



Kiwi



Strawberry



Corn

Some seeds we eat as food. They are rice, wheat, maize, pulses, beans, grains, etc.



Rice



Wheat



Gram

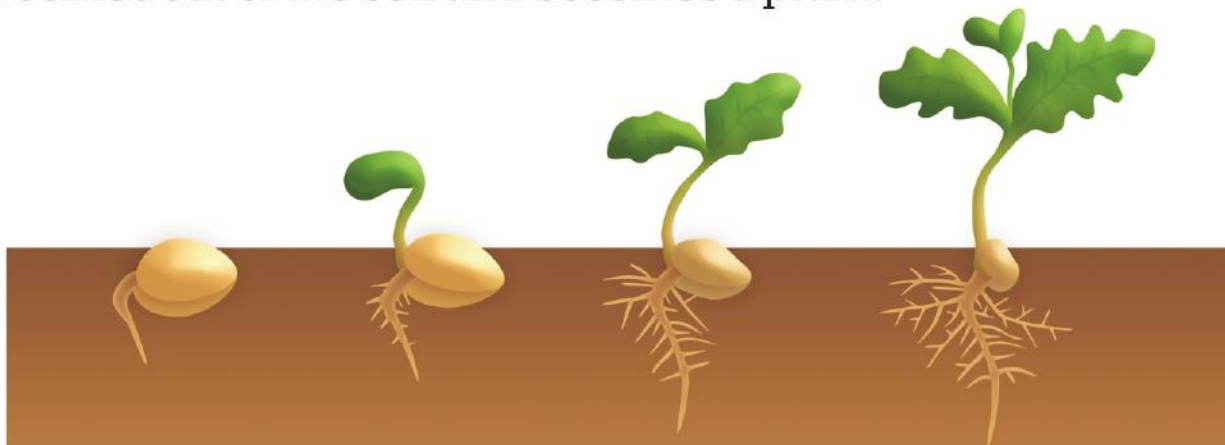


Pulses

GROWING OF A SEED INTO A PLANT

A plant grows from a seed. A seed has a baby plant inside it. To become a plant, a seed needs soil, air, water and sunlight in the right amount.

We sow a seed in the moist soil. The sunlight keeps the seed warm. Air in the soil helps it to breathe. The seed becomes soft and after some days it starts to germinate. The baby plant which resides in the seed comes out of the soil and becomes a plant.



The sunlight gives warmth to the seed. Water provides hydration to the seed and makes it soft. Air helps to breathe and soil provides minerals to the seed. When a seed becomes a plant, its leaves make food for the plant.



Not all plants grow from seeds. Some plants grow from roots, stems and leaves.

KEY WORDS

Seed	—	hard thing inside the fruit
Reside	—	to live
Minerals	—	useful things needed by a plant
Germinate	—	to grow



Backpedal

- ▶ Most of the plants grow from seeds.
- ▶ We do not eat seeds of some plants.
- ▶ Some seeds we eat as food.
- ▶ A seed has a baby plant inside it.
- ▶ A seed needs sunlight (warmth), water, air and soil to become a plant.



TESTING TIME



Objective Type Questions

Continuous Review

A. Tick (✓) the correct option:

1. Plants grow from _____.

(i) flowers



(ii) fruits



(iii) seeds



2. Flowers have _____ in them.
(i) fruits ☐ (ii) seeds ☐ (iii) None of these ☐
3. A seed is sown in the _____.
(i) sand ☐ (ii) cement ☐ (iii) soil ☐

B. Fill in the blanks with correct words:

1. Most of the flowers bear _____.
2. Not all flowers have _____.
3. Some fruits have only _____ seed.
4. A seed has a _____ inside it.

one
baby plant
fragrance
fruits

C. Write True or False:

1. Flowers are used in perfumes, soaps, etc.
2. All fruits have seeds inside them.
3. There are some seeds that can be eaten.
4. Seeds need soil, warmth, air and water to grow.

Subjective Type Questions

A. Answer in one word:

1. Plants grow from this. _____
2. Fruits have these. _____
3. This fruit has only one seed. _____
4. Warmth, air, water and soil are necessary for this to grow. _____

B. Answer the following questions:

1. How are flowers used?

2. What is needed to grow a plant?

3. What does a seed need to grow?

4. Where does the baby plant reside?

Think and Respond

- Can a plant grow without a seed? How?

Critical Thinking

Life skills and Values



Environmental Literacy

- How can you help a seed to grow?

FUNNY PROJECTS



- Collect some seeds. Put them in a pouch and paste them in your scrapbook. Write their names too.
- Name the given seeds that we eat as food.

Experiential Learning

Recognising Skills



- Draw a seed and a plant in your notebook.

Art Integration



4

FOOD FROM PLANTS



LEARNING OUTCOMES

- ▶ Fruits
- ▶ Vegetables
- ▶ Grains
- ▶ Pulses
- ▶ Spices
- ▶ Nuts
- ▶ Beverages
- ▶ Sugar
- ▶ Oil seeds

We have learnt that a seed grows into a plant. Plants give us fruits, vegetables, grains, pulses, etc. These things make our food. Thus, plants give us food.

FRUITS

We eat fruits. Fruits are juicy and fleshy part of a plant. They can be sweet like mango or sour like lemon. Some fruits are apple, banana, pomegranate, guava, etc.



Apple



Mango



Banana



Guava



Pomegranate

VEGETABLES

We get vegetables from plants. We eat them in the form of root, stem, leaf, flower or fruit. They are also fleshy, but not juicy. Some vegetables are potato, onion, brinjal, cauliflower, lady's finger, etc.



Onion



Potato



Brinjal



Lady's finger



Cauliflower



Cucumber is a fruit not a vegetable.

GRAINS

Cereals are the grains that we eat. We get cereals from plants. They are rice, wheat, corn, barley, etc.



Rice



Wheat



Corn



Barley



Pulses

PULSES

Pulses are also seeds. They are an important part of our food. They are arhar, moong, chana daal, malka, etc.



Arhar daal



Moong daal



Chana daal



Malka daal



Urad daal



Beans are pulses.

Click Your Brain

Identify these pictures.



SPICES

We get spices from plants. They add flavour and taste to our food. They are cumin, turmeric, red chillies, clove, cardamom, etc.



Turmeric



Cumin



Red Chillies



Cardamom



Cloves



Spices have medicinal values also.

NUTS

Nuts are dry fruits. They are oily. They are almond, cashewnut, raisin, pistachio, dates, walnut, groundnut, etc.



Almond



Cashewnut



Raisin



Pistachio



Groundnut

BEVERAGES

We drink beverages. Tea and coffee are beverages. Tea is obtained from tea plant leaves and coffee is obtained from the beans of the cocoa plant.



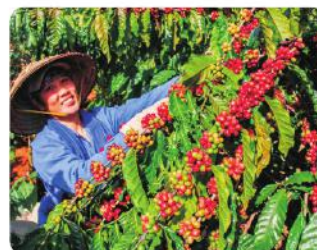
Tea



Tea Garden



Coffee



Coffee Garden

SUGAR

Sugar is obtained from sugarcane plant.



Sugarcane

OIL SEEDS

We get oil from seeds, like mustard, sunflower, etc.



Mustard Oil



Sunflower Oil

KEY WORDS

Grains	—	seeds
Pulses	—	include beans, lentils and peas
Spices	—	food-items that add taste and flavour to our food
Nuts	—	dry fruits
Beverage	—	drinkable food items



Backpedal

- ▶ Fruits have seeds inside them.
- ▶ Plants grow from seeds.
- ▶ We get fruits, vegetables, grains, pulses, beverages, etc. from plants.



TESTING TIME



Objective Type Questions

Continuous Review

A. Tick (✓) the correct option:

- Fruits are _____.
(i) juicy ☐ (ii) fleshy ☐ (iii) Both of these ☐
- Cucumber is a _____.
(i) fruit ☐ (ii) vegetable ☐ (iii) grain ☐
- Spices add _____.
(i) flavour ☐ (ii) taste ☐ (iii) Both of these ☐

B. Fill in the blanks with correct words:

- Plants give us _____.
- Grains are called _____.
- _____ is a pulse.
- Tea and coffee are _____.

Arhar
beverages
cereals
food

C. Write True or False:

- Plants give us many things.
- Vegetables are grains.
- Rice and wheat are cereals.
- Sugar is obtained from sugarcane.

Subjective Type Questions

A. Answer in one word:

1. These are juicy and fleshy.
2. These are fleshy but not juicy.
3. These have medicinal values also.

B. Answer the following questions:

1. What are the things that we get from plants?

2. What are fruits?

3. Name some pulses and cereals.

4. What do the spices do to our food?

Think and Respond

- There are some fruits which are cooked as vegetables. Write their names.

Critical Thinking

Life skills and Values



Social Literacy

- Many a time, you must have left your food and have thrown it. Should you do this whereas there are so many children who do not have food?

FUNNY PROJECTS



- Make a list of food-items you eat in your different meals.

Experiential Learning



5

ANIMAL KINGDOM



LEARNING OUTCOMES

- ▶ Land Animals
- ▶ Water Animals
- ▶ Land and Water Animals
- ▶ Birds
- ▶ Arboreal Animals
- ▶ Insects
- ▶ Worms

We have learnt about living things. Animals are also living things. We will study about animals in this chapter.

You can see animals around you. They are living beings. They move, breathe, eat food, drink water, reproduce and grow. Animals can be small or big. They live on land, in water or on the trees.

LAND ANIMALS

Animals that live on land are called **land animals**. Examples: dog, cow, lion, giraffe, zebra, elephant and horse, etc.



Cow



Horse



Lion



Giraffe



Zebra

We can divide land animals as wild animals, domestic animals and pet animals.

Wild Animals : Animals that live in forests are called **wild animals**. They are very dangerous and fierce. Examples : lion, tiger, giraffe, zebra, rhinoceros, etc. You can see wild animals in a zoo or sanctuary.



Lion



Tiger



Giraffe



Rhinoceros



Zebra

Domestic Animals : Humans tame some animals for their use. They are called **domestic animals**. They are buffalo, cow, horse, mule, sheep, camel, etc.



Cow



Buffalo



Horse



Mule



Sheep

Pet Animals : Animals that are kept by humans for their company and enjoyment are called **pet animals**. They are dog, cat, rabbit, pig, parrot, etc.



Dog



Cat



Rabbit



Parrot

WATER ANIMALS

Some animals live in water. They are called **water animals**. We call them as aquatic animals also. They are fish, dolphin, shark, whale, octopus, starfish, etc.



Starfish



Shark



Dolphin



Whale



Octopus

LAND AND WATER ANIMALS

There are some animals that live both on land and in water. They are frog, crocodile, tortoise, hippopotamus, etc.



Frog



Crocodile



Tortoise



Hippopotamus



Animals that live both on land and in water are called **amphibians**.

BIRDS

Animals that fly are called birds. They have two wings to fly. They have two legs to walk. Crow, parrot, sparrow, kingfisher, pigeon, owl, kite, etc. are birds. We call them as **aerial animals**.



Parrot



Pigeon



Owl



Sparrow



Kite

Click Your Brain

Which kind of animals are:

1. Frog _____
3. Parrot _____
5. Cow _____

2. Dog _____
4. Lion _____
6. Shark _____



1. Hen and peacock are also birds but they cannot fly high.
2. Kiwi and ostrich are flightless birds.
3. Duck can swim.

ARBOREAL ANIMALS

There are some animals that spend most of their time on trees. They are **arboreal animals**. Examples : monkey, sloth, koala, squirrel, etc.



Monkey



Sloth



Koala



Squirrel

INSECTS

Some animals are very small. They are called **insects**. They have six legs. Most insects have wings. Examples: Ant, housefly, spider, bee, beetle, cockroach, cricket, butterfly, etc. are **insects**.



Spider



Ant



Butterfly



Cricket



Bee



1. Butterfly has four colourful wings.
2. Louse has no wings.

WORMS

Worms are boneless animals. They can only crawl on land. They are earthworm, centepede, leech, nercis, bristle worm, etc.



Earthworm



Centepede



Leech



Bristle



Nercis

Click Your Brain

Give examples:

Arboreal animals	_____	_____
Insects	_____	_____
Worms	_____	_____



Snakes, lizards, crocodiles are reptiles. They have scales on their bodies.

KEY WORDS

Aquatic animals	—	animals that live in water
Amphibians	—	animals that can live both on land and in water
Arboreal animals	—	animals that live on trees
Aerial animals	—	animals that spend most of their time in the air
Insects	—	very small animals
Worms	—	boneless animals
Flightless	—	that cannot fly



Backpedal

- ▶ Animals are divided as land animals, aerial animals and water animals.
- ▶ Land animals can be wild, domestic and pet.
- ▶ Arboreal animals spend most of their time on trees.
- ▶ Insects are very small animals with wings and six legs.
- ▶ Worms are boneless animals.



TESTING TIME



Objective Type Questions

Continuous Review

A. Tick (✓) the correct option:

1. Animals generally have _____ legs.

(i) two

☐

(ii) three

☐

(iii) four

☐

2. A lion is a _____ animal.

(i) land

☐

(ii) water

☐

(iii) air

☐

3. Birds _____.

(i) crawl

☐

(ii) walk

☐

(iii) fly

☐

B. Fill in the blanks with correct words:

1. _____ animals live in jungle.

2. _____ is a domestic animal.

3. An octopus lives in _____.

4. Monkeys live on _____.

Horse
water
trees
Wild

C. Write True or False:

1. Land animals live on land.

2. Fish have wings.

3. Insects have six legs.

4. Earthworm is a water animal.

Subjective Type Questions

A. Answer in one word:

1. What are the animals called that live in water? _____
2. What type of animal is a squirrel? _____
3. Who crawls? _____
4. Name one worm and one insect. _____

B. Answer the following questions:

1. What are land animals? What are their types?

2. What are water animals? Give examples.

3. What are birds? Give examples.

4. Which animals live on trees?

Think and Respond

- How are animals useful for us?

Critical Thinking

Life skills and Values

Social Literacy

- How should we behave with animals?

FUNNY PROJECTS

- Make a list of animals you see around yourself in daily life. Experimental Learning



6

TYPES OF ANIMALS



LEARNING OUTCOMES

► Food

► Shelter or House

Every living thing has some basic needs. They are food, shelter and clothes. Animals also need food and shelter.

FOOD

Food is the primary need to grow and live. It gives energy and nutrition. Animals eat different kinds of foods. They eat grass, plants, leaves, fruits, seeds, and flesh of other animals. On the basis of food they eat, animals can be herbivores, carnivores, omnivores or scavengers.

Herbivores : Some animals eat plant products like grass, leaves, fruits, seeds, grains, etc. These animals are called **herbivores**. They are cow, horse, goat, camel, elephant, giraffe, etc.



Cow



Horse



Goat



Elephant



Camel

Carnivores : Some animals eat only flesh of other animals. They are called **carnivores**. They are lion, tiger, crocodile, fox, leopard, etc.



Lion



Tiger



Fox



Crocodile



Leopard

Omnivores : Some animals eat both plant products and flesh of other animals. They are called **omnivores**. They are dog, cat, crow, bear, mouse, kiwi, etc.



Dog



Cat



Crow



Bear



Kiwi

Scavengers : Animals that eat dead animals are called **scavengers**. They are vulture, jackal, coyote, piranha fish, raccoon, etc.



Vulture



Jackal



Coyote



Piranha



Raccoon

SHELTER OR HOUSE

Animals also need shelter. They need it to lay eggs and for protection. Some animals make their shelter themselves. Some of them are:



A lion lives in a den.



A Bird lives in a nest.



A rabbit lives in a burrow.



A snake lives in a hole.



A spider lives in a web.



Bees live in a beehive.



Ants live in an ant colony.



Elephant and giraffe live under trees.

Humans also make shelters for animals.



A cow lives in a shed.



A dog lives in a kennel.



A horse lives in a stable.



Hens live in a coop.

Click Your Brain

Name these.

1. Herbivores
2. Carnivores
3. Omnivores
4. Scavengers
5. Birds build it
6. Lions live here
7. Horses lives here
8. Hens live here

_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____

KEY WORDS

Herbivores — plant-eating animals
Carnivores — flesh-eating animals

Omnivores	—	plant and flesh-eaters
Scavengers	—	dead animal eaters
Shelter	—	place for living and protection
Flesh	—	soft part of animals or humans under the skin



Backpedal

- ▶ Animals need food and shelter.
- ▶ Herbivores are plant-eaters.
- ▶ Carnivores are flesh-eaters.
- ▶ Omnivores are plant and flesh-eaters.
- ▶ Scavengers eat dead animals.
- ▶ A shelter is needed for protection and to lay eggs.



TESTING TIME



Objective Type Questions

Continuous Review

A. Tick (✓) the correct option:

- The basic need of animals is _____.
 (i) food ☐ (ii) shelter ☐ (iii) Both of these ☐
- A horse is a _____.
 (i) herbivore ☐ (ii) carnivore ☐ (iii) omnivore ☐
- _____ eat dead animals.
 (i) Carnivores ☐ (ii) Omnivores ☐ (iii) Scavengers ☐

B. Fill in the blanks with correct words:

- Food and shelter are _____ needs.
- _____ are omnivores.
- A house gives _____.
- A cow lives in a _____.

protection
shed
Crows
basic

C. Write True or False:

1. All living things need food and shelter.
2. A carnivore eats grass.
3. Bees live in a beehive.
4. A horse lives in a kennel.

Subjective Type Questions

A. Answer in one word:

1. They eat grass.
2. They eat flesh of other animals.
3. They eat both grass and flesh.
4. They eat dead animals.

B. Answer the following questions:

1. What are basic needs?

2. Why do animals need food and shelter?

3. Differentiate between herbivores and carnivores.

Think and Respond

- Why do different animals eat different foods? **Critical Thinking**

Life skills and Values



Environmental Literacy

- We are destroying our trees and forests. How can we maintain our earth green?

FUNNY PROJECTS



- Paste the pictures of some wild animals in your scrapbook.
- Draw a den and a spider's web in your scrapbook.

Curiosity

Art Integration



7

OUR BODY



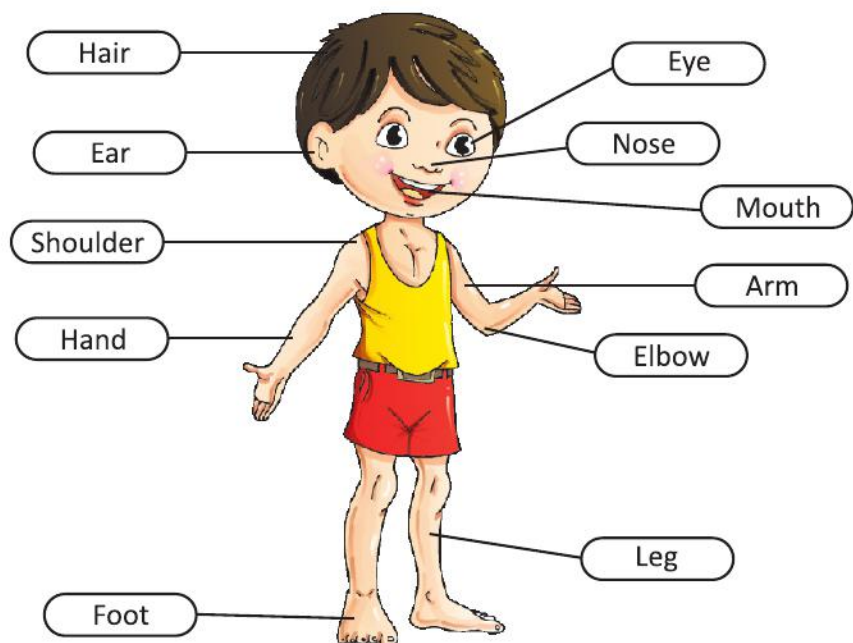
LEARNING OUTCOMES

- ▶ Outer Body Parts
- ▶ Inner Body Parts
- ▶ Uses of Body Parts
- ▶ Sense Organs

You have learnt about the parts of plants. Let us learn about the parts of our body.

Our body has different parts. Some parts are outside of our body. They are called **outer body parts**. They are eyes, nose, ears, hands, legs, feet, teeth, fingers, etc. Some parts are inside of our body. They are called **inner body parts**. They are heart, lungs, kidneys, intestines, etc. Each part of our body is important.

OUTER BODY PARTS



Parts of Human Body



You can touch your outer body parts.

INNER BODY PARTS



Heart



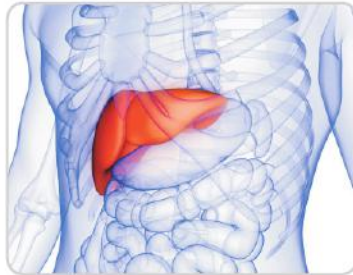
Lungs



Intestines



Kidneys



Liver



Brain



We cannot touch our inner body parts.

USES OF BODY PARTS

We use our outer body parts differently.

Hands : Hands hold, push or pull things and write.

Legs : We use legs to walk, run, kick, etc.

Mouth : Mouth helps to eat and speak.

Teeth : Teeth cut and chew food.

Likewise other body parts have different uses.

Our inner body parts have the following uses:

Brain : Brain thinks and controls the functions of the body.

- Heart** : Heart pumps blood to the whole body.
- Lungs** : Lungs help in breathing.
- Kidneys** : Kidneys remove wastes out of our body.
- Intestines** : Intestines absorb important materials of food and separate the waste.
- Liver** : The liver helps in digestion.

Click Your Brain

Give examples :

1. Outer body parts _____
2. Inner body parts _____

SENSE ORGANS

Sense organs help us to know our surroundings. They are eyes, ears, nose, skin and tongue.

Eyes help us to see. **Ears** help us to hear or listen.

Nose helps us to smell good or bad odour.

We feel with our **skin**. It tells us things as hot, cold, rough, smooth, etc.

We taste with our **tongue**. It tells us about sweet, sour or salty things.



Eyes



Ears



Nose



Skin



Tongue

Click Your Brain

Complete the following.

1. We hear with _____.
2. We see with _____.
3. We smell with _____.
4. We feel with _____.
5. We taste with _____.

KEY WORDS

Outer — outside

Inner — inside

Organs — body parts

Odour — smell



Backpedal

- ▶ Our body has different parts.
- ▶ We can touch our outer body parts.
- ▶ We cannot touch our inner body parts.
- ▶ We have five sense organs— eyes, ears, nose, tongue and skin.
- ▶ Every body part has a different work to do.
- ▶ Each part of our body is important.



TESTING TIME



Objective Type Questions

Continuous Review

A. Tick (✓) the correct option:

1. We have different _____ parts.

(i) body



(ii) machine



(iii) outer



2. We can touch our _____ body parts.

(i) outer

(ii) inner

(iii) upper

3. We have _____ sense organs.

(i) four

(ii) five

(iii) six

B. Fill in the blanks with correct words:

1. Eyes, ears, nose, etc. are _____ body parts.

2. We cannot touch our _____ body parts.

3. The _____ thinks and controls the body.

4. We _____ with our skin.

inner

brain

feel

outer

C. Write True or False:

1. Some body parts are outside of our body.

2. Hands hold, push or pull things.

3. Kidneys help in digestion.

4. Nose helps us to smell good or bad odour.

Subjective Type Questions

A. Answer in one word:

1. What do we call the body parts that we can touch? _____

2. Which body parts cannot be touched? _____

3. How many sense organs are there in our body? _____

4. What helps us to feel? _____

B. Answer the following questions:

1. What are outer and inner body parts?

2. Name some inner body parts.

3. Write the uses of any two outer body parts.

4. Name all the sense organs.

Think and Respond

- How are the sense organs different from other outer body parts?

Critical Thinking

Life skills and Values



Healthy Literacy

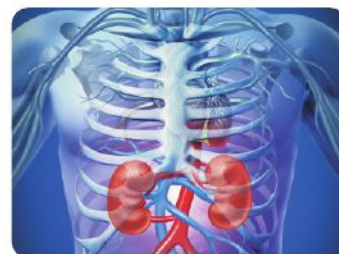
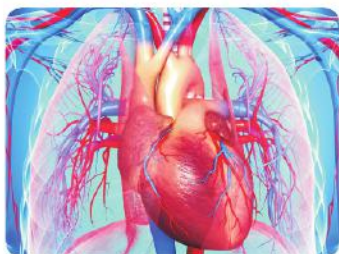
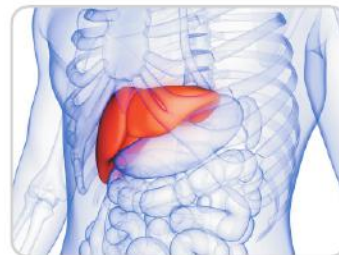
- How can we take care of our body parts?

FUNNY PROJECTS



- Name the given body parts.

Recognising Skills



- Draw the pictures of sense organs in your notebook.

Art Integration



8

STAYING HEALTHY



LEARNING OUTCOMES

► Cleanliness

► Keeping Fit

In the previous chapter, you have learnt about different body parts. In this chapter, we shall teach you about some habits which keep you healthy and fit. To be healthy, you must follow cleanliness.

CLEANLINESS

Cleanliness means to be free from germs, dirt and waste. To maintain cleanliness, follow the given schedule:

- Take bath everyday.
- Brush your teeth twice a day. Use soft brush to clean your teeth.
- Wash your hair with mild shampoo.
- Go to washroom daily.
- Oil your hair when they are dry.
- Wash your hands as to needed.



- Trim your nails regularly.
- Wash your eyes with fresh water.
- Keep your nose clean. Never put your finger into your nose.
- Cover your mouth while sneezing and coughing.
- Do not bite your nails.
- Use only your towel and comb.



Germes are tiny; we cannot see them. So cleanliness is necessary.

Click Your Brain

Write some ways to keep yourself clean.

Tick (✓) the activity, which is right.



KEEPING FIT

We also do some activities to keep ourselves fit. Here are some of them.

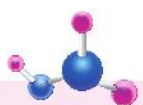
- Go to bed early at night and get up early in the morning.

- Eat healthy food slowly and chew it well. Drink plenty of clean water.
- Maintain correct posture while walking or sitting.
- Breathe through your nose.
- Do some light **yoga asanas**.
- Do **Pranayam**.
- Go for a walk every morning.
- Play any game or go for cycling.
- Do some exercises.
- Eat healthy food on time.
- Do not spend too much time on mobile.
- Never cover your face while sleeping.



KEY WORDS

Cleanliness	—	state of being clean
Pranayam	—	a yoga pose
Asanas	—	sitting meditation poses



Backpedal

- ▶ To be healthy and fit, we should maintain cleanliness.
- ▶ We should do some exercises, yoga, pranayam, etc. to be fit and healthy.
- ▶ Eat healthy food slowly and chew it well. Drink plenty of clean water.

Objective Type Questions

Continuous Review

A. Tick (✓) the correct option:

- To be healthy, you must follow _____.
 (i) unhygienic habits ☐ (ii) bad habits ☐
 (iii) cleanliness ☐
- Take a bath _____.
 (i) daily ☐ (ii) once a week ☐
 (iii) sometimes ☐
- _____ makes us fit.
 (i) Exercise ☐ (ii) Yoga ☐ (iii) Both of these ☐

B. Fill in the blanks with correct words:

- Brush your teeth _____ a day.
- Trim your _____ regularly.
- Do not _____ your nails.
- Drink _____ of clean water.

plenty
bite
twice
nails

C. Write True or False:

- Use soft brush to clean your teeth.
- Wash your eyes with warm water.
- Go to bed early at night and get up early in the morning.
- Do not do the exercise.

Subjective Type Questions

A. Answer in one word:

- You should trim these. _____
- You should do this before and after having food. _____
- Breathe through this. _____

4. You should eat this to be fit and healthy. _____

B. Answer the following questions:

1. What is the meaning of cleanliness?

2. What should we do to keep our body clean?

3. What activities should we do to be fit?

4. Write two ways of cleanliness and being healthy?

Think and Respond

- Why is it important to be fit and healthy?

Critical Thinking

Life skills and Values



Healthy Literacy

- What type of activities should we do to be fit and healthy?

FUNNY PROJECTS



- Identify the pictures and write the names of these activities. **Recognising Skills**









- Paste the pictures of some healthy activities in your scrapbook. **Art Integration**



9

HOUSES AND CLOTHES



LEARNING OUTCOMES

- ▶ Need for a House
- ▶ Types of Houses
- ▶ Some Other Houses
- ▶ A Good House
- ▶ Clothes

All living beings need a house to live in. It is the basic need of living beings.

It protects us from heat, cold, rain, storm, thieves and wild animals.

NEED FOR A HOUSE

House is a place where we live in. It keeps us safe.

TYPES OF HOUSES

Mainly houses are of two types — kutchha house and pucca house .

Kutchha House : A kutchha house is made up of mud, straw, bamboo, etc. Kutchha houses are mostly found in villages. They are not much strong.



Kutchha House



Pucca House

Pucca House : A pucca house is made of cement, bricks, iron, etc. Pucca houses are strong and found mostly in cities. Nowadays, you can see them in villages too.

SOME OTHER HOUSES

Some other houses are tent, houseboat, bungalow, caravan, igloo, etc.



Tent



Houseboat



Bungalow



Flats



Caravan



Stilt House



Igloo



Stone House



An Igloo is a house made of snow blocks.

A GOOD HOUSE

- A good house is airy.
- It has good ventilation.
- There are high walls in a good house.
- There is net on windows and doors.
- A good house is spacious. It has bedroom, drawing room, dining room, study room, kitchen, storeroom, bathroom and toilet.
- A good house has good drainage system.

Click Your Brain

Look at the pictures given below and write kutcha house or pucca house.



CLOTHES

We need clothes to cover our body. Clothes protect us from heat, cold, dust, rain and germs.

There are different types of clothes, like cotton clothes, woollen clothes, silk clothes, etc.

We wear clothes according to seasons.

We wear light-coloured cotton clothes in summer. They keep us cool.



Clothes in Summer



Clothes in Rainy Season

We wear synthetic and cotton clothes in rainy season. We use raincoats, umbrellas, gumboots to protect ourselves from being wet.

We wear woollen clothes in winter. They keep us warm and protect us from cold.



Clothes in Winter

KEY WORDS

- Ventilation — a passage to let go the warm air out
 Drainage system — a system to let go the dirty water out



Backpedal

- ▶ A house protects us from heat, cold, rain and thieves.
- ▶ A house can be kutcha house or pucca house.
- ▶ Some other houses are houseboat, igloo, tent, caravan, bungalow, stilt house and flat.
- ▶ Clothes protect us from heat, cold, rain and germs.
- ▶ We wear clothes according to seasons.



Objective Type Questions

Continuous Review

A. Tick (✓) the correct option:

1. A house _____ us.
 (i) harms ☐ (ii) protects ☐ (iii) troubles ☐
2. A house can be _____.
 (i) kutcha ☐ (ii) pucca ☐ (iii) Both of these ☐
3. We wear a raincoat in _____ season.
 (i) summer ☐ (ii) winter ☐ (iii) rainy ☐

B. Fill in the blanks with correct words:

1. A house protects us from heat, cold, _____ and thieves.
2. A pucca house is _____.
3. We need clothes to _____ our body.

strong
rain
cover

C. Write True or False:

1. A kutcha house can be seen in a village.
2. A good house does not have ventilation.
3. We wear only one type of clothes.

Subjective Type Questions

A. Answer in one word:

1. These houses are found in cities.
2. These houses are found in villages.
3. We wear these clothes in summer.

B. Answer the following questions:

1. Why do we need a house?
2. How many kinds of houses are there?
3. Why do we need clothes?

Think and Respond

- Why are house and clothes important for us? **Critical Thinking**

Life skills and Values



Environmental Literacy

- Why do some houses have flat roofs and some have slopy roofs?

FUNNY PROJECTS



- Make a model of an igloo with thermocol.

Creativity



10

WATER : A PRECIOUS GIFT



LEARNING OUTCOMES

- ▶ Need of Water
- ▶ Sources of Water
- ▶ Uses of Water
- ▶ Clean Water

Water is precious. We cannot live without water. About 70% area of our earth is covered with water. A normal human body also contains 70% of water. Humans, animals, birds, plants, all need water.

NEED OF WATER

Everyone needs water to drink. Water keeps the body cool. It helps in digestion.

USES OF WATER

We use water for:

- washing clothes.
- cleaning floor, utensils, etc.
- cooking food.



- putting off fire.
- flushing toilets.
- bathing.
- watering plants and fields.



You get water from taps. But there are many places where people have to go very far-off places to fetch water.

SOURCES OF WATER

The main source of water is the rain. It is the purest form of water.



Rain



River



Pond



Lake



Well

Rivers, lakes and ponds are some other sources of water. We also get water from wells. Oceans and seas also have water. But it is not fit for drinking.



Water in oceans and seas is salty.

Click Your Brain

Complete the following.

1. We need water to _____.
2. We use water for _____, _____, _____, etc.

3. The main source of water is _____.
4. Some other sources of water are _____, _____, _____.
5. There is salty water in _____ and _____.

CLEAN WATER

We should drink clean water. If we drink dirty water, we can fall ill. We can make the dirty water clean by:

- boiling.
- filtering.
- putting chlorine into water.



Waste and germs make the water impure.

KEY WORDS

Utensils — pots and pans used in the kitchen



Backpedal

- ▶ We need water to drink.
- ▶ We use water for washing, cleaning, bathing, etc.
- ▶ The main source of water is the rain.
- ▶ We get water from rivers, ponds, lakes and wells.
- ▶ Oceans and seas have salty water.

Objective Type Questions

Continuous Review

A. Tick (✓) the correct option:

- We need water to _____.
 (i) drink ☐ (ii) bath ☐ (iii) Both of these ☐
- The main source of water is the _____.
 (i) river ☐ (ii) pond ☐ (iii) rain ☐
- Oceans and seas have _____ water.
 (i) sweet ☐ (ii) bitter ☐ (iii) salty ☐

B. Fill in the blanks with correct words:

- We cannot live without _____.
- Water is used in _____ the plants.
- Water helps in _____.
- Water in oceans and seas is not _____ for drinking.

watering
digestion
fit
water

C. Write True or False:

- About 70% area of our earth is covered with water.
- Only humans need water.
- Water keeps our body cool.
- We can clean water by boiling it.

Subjective Type Questions

A. Answer in one word:

- How much water is there on the earth? _____
- It is the purest form of water. _____
- It may make us ill. _____
- We can clean water by doing this. _____

B. Answer the following questions:

1. Why do we need water?

2. What are the uses of water?

3. What is the main source of water?

4. What are the methods to make the water clean?

Think and Respond

- Why is it not good to drink dirty water?

Critical Thinking

Life skills and Values



Environmental Literacy

- Should we waste water or not? Why?

FUNNY PROJECTS



- Write a paragraph on 'Water'.

Curiosity

- Make a water drop on a drawing sheet and colour it.

Art Integration



11

AIR : OUR LIFELINE



LEARNING OUTCOMES

► Uses of Air

► Qualities of Air

We cannot live without air. Air is one of our basic needs. Every living being needs air to breathe and to be alive.

There is no life without air. Air has oxygen which we breathe in and carbon dioxide which plants breathe in.

USES OF AIR

Air is used for:

- breathing.
- burning fire.
- blowing balloons.
- gliding hot air balloons.
- flying kites.
- drying clothes.



Breathing



Flying Kites



Blowing Balloons



Drying Clothes

QUALITIES OF AIR

Air has the following qualities:

- We cannot see air, we can feel it.
- Air is colourless.
- Air occupies space.
- Air has force.
- Air has weight.
- Air has mass.
- Air exerts pressure.



Air

ACTIVITY

Air Helps in Burning.

Take two candles, two glasses and one lid. Light both the candles. Put each candle under each glass. Now, cover one glass with the lid. After some time, the candle kept in the glass with lid on it, blows out and the other candle keeps on burning. It proves that the fire needs air.



Air Occupies Space.

Take one glass and a half filled tub with water. Try to put the glass into water straight. It will not get into the water. Now, tilt the glass. Push the glass into the water again. Now it can go into the water. It proves that air occupies space.

KEY WORDS

- Breathing — process of taking the air in and out
Gliding — move smoothly and easily without power



Backpedal

- ▶ Air is very important.
- ▶ Air is needed to breathe.

- ▶ We need air for burning fire, flying kites, etc.
- ▶ Air has many qualities.



Objective Type Questions

Continuous Review

A. Tick (✓) the correct option:

- | | | | |
|-----------------------------------|--------------------------|---------------|--------------------------|
| 1. We cannot live without _____. | | | |
| (i) pen | ☐ | (ii) paper | ☐ |
| | | (iii) air | ☐ |
| 2. We need air to _____. | | | |
| (i) see | ☐ | (ii) eat | ☐ |
| | | (iii) breathe | ☐ |
| 3. Air is used for burning _____. | | | |
| (i) paper | ☐ | (ii) fire | ☐ |
| | | (iii) cloth | ☐ |

B. Fill in the blanks with correct words:

- We need air to _____.
- Air has _____.
- Air is _____.
- Air occupies _____.

colourless
space
oxygen
breathe

C. Write True or False:

- Every living being needs air.
- Plants breathe in oxygen.
- Air helps in flying kites.
- Air has no mass.

Subjective Type Questions

A. Answer in one word:

- What do we need to be alive?

2. Air exerts this.
3. Air occupy this.
4. Plants breathe in this.

B. Answer the following questions:

1. Why do we need air?

2. What do we breathe in?

3. Write three uses of air.

4. What are the qualities of air?

Think and Respond

- Do water animals breathe? How?

Critical Thinking

Life skills and Values



Environmental Literacy

- How are we responsible to make the air dirty?

FUNNY PROJECTS



- Write some uses of air other than discussed in the chapter. **Experiential Learning**

- Draw a picture showing the hot air balloon in your notebook.

Art Integration



12

OUR PLANET EARTH



LEARNING OUTCOMES

► The Earth

► The Sun

► The Moon

► The Stars

THE EARTH

We live on the Earth. It is our home. It is the third planet in the solar system. It is round in shape. It is called the **blue planet** because of water on its surface.

There are plants and trees on the Earth.

It moves around the Sun and on its axis. It completes one round around the Sun in every 24 hours. We call this time period as **one day**.



Earth



The Earth is the only planet to have life on it.

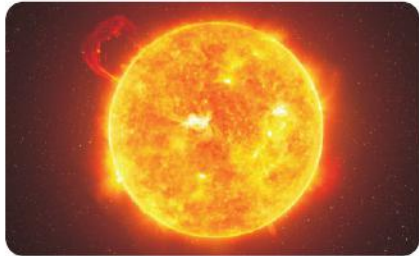
THE SUN

The Sun is a star. It is very big. It is a huge ball of fire. It is very hot. It gives us heat and light. It is very far from the Earth. It brings day and night on the Earth.

The Sun rises in the east and sets in the west.

The rays of the Sun produce heat. Heat is very important for the growth of plants.

There will be no light and heat in the absence of the Sun.



Sun



Rising Sun



Setting Sun

THE MOON

The Moon is a satellite of the Earth. It does not have its own light. It shines bright with the light of the Sun.

You can see it in the night sky.

It moves around the Earth. It changes its shape everyday. When it is a complete circle, it is called the **full moon**. When there is no moon, it is called the **new moon**. New moon generally cannot be seen.



Moon



Phases of the Moon

THE STARS

We see stars in the sky at night.

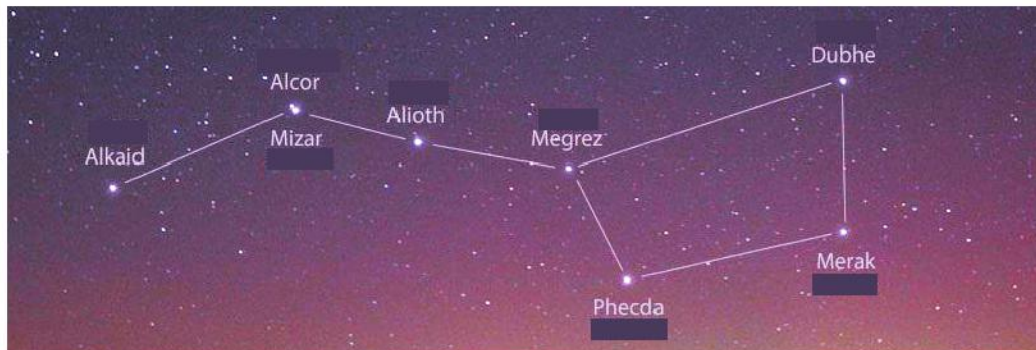
They are bright and twinkling. They are of different sizes. The Sun is also a star. Stars are stationary. They have their own light.

They look so small because they are very far from us. Some stars are bigger than the Sun.

There are some stars which form some shape. These are called **constellation**. Example – **Saptarishi**.



Stars



Saptarishi

Click Your Brain

1. What is the Sun?

2. What is the Earth?

3. What is the Moon?



Stars are there in the day time, but in the Sun's bright light they are not visible to us.

KEY WORDS

Planet — a body which revolves around the Sun.
Stationary — fixed



Backpedal

- ▶ The Earth is a planet. We live on it.
- ▶ The Sun is a huge ball of fire.
- ▶ The Moon is a satellite of the Earth.
- ▶ We see the stars at night.



TESTING TIME



Objective Type Questions

Continuous Review

A. Tick (✓) the correct option:

1. We live on the _____.

(i) Earth

☐

(ii) Moon

☐

(iii) stars

☐

2. The earth moves around the _____.

(i) Moon

☐

(ii) Sun

☐

(iii) stars

☐

3. The _____ is the satellite of the Earth.

(i) Sun

☐

(ii) Moon

☐

(iii) stars

☐

B. Fill in the blanks with correct words:

1. The Earth gets light from the _____.

2. The Sun is a _____.

3. The Sun rises in the _____.

4. The Moon is a _____ of the Earth.

satellite

east

star

Sun

C. Write True or False:

1. The Earth moves around its axis.

2. The Sun is very near to us.

3. The Moon moves around the Sun.

4. The stars are stationary.

Subjective Type Questions

A. Answer in one word:

1. What is the Earth?
2. What is the Sun?
3. What is the Moon?
4. What do we see at night in the sky?

B. Answer the following questions:

1. Why is the Earth called the blue planet?

2. What do we get from the Sun?

3. What is the full moon?

4. What is the Saptarishi?

Think and Respond

- How do we get light?

Critical Thinking

Life skills and Values



Environmental Literacy

- The Earth moves. But we do not fall. How?

FUNNY PROJECTS



- Write the names of other planets that move around the Sun.
- Draw the picture of the Earth with the Sun and the Moon in your notebook.

Curiosity

Art Integration



13

WEATHER AND SEASONS



LEARNING OUTCOMES

► Weather

► Seasons

There is a blanket of gases all around the earth. It is called the **atmosphere**. The air, heat and water present on the earth, make its conditions cool, hot, windy, etc. on the earth.

WEATHER

When one morning you wake up, it is cool, by the noon it becomes hot, and in the evening it becomes pleasant and starts to rain. These changing conditions of atmosphere are called as **weather**.



Cool Morning



Hot Noon



Rainy Evening

The weather changes hour to hour or day to day. When the sun is strong, it is **sunny**; when air is strong, it is **windy**; when there are

clouds in the sky, it is **cloudy**; when the clouds pour down water, it is **rainy**.



Sunny day



Windy day



Cloudy day



Rainy day

SEASONS

When the weather remains the same for a long time, we call it **season**. There are three main seasons in India. They are:

- Summer Season
- Winter Season
- Rainy (Monsoon) Season

Summer Season : Summer season is hot and dry. The sun shines brightly. It is harsh at noon. People wear light-coloured cotton clothes. They like to stay indoors. They enjoy cold drinks that make the atmosphere cool. They sit in AC rooms or in front of the coolers.



Summer Season

Warm winds blow during this season. They are called **Loo**. Many people go to hill stations to escape from the hot season.

Rainy Season : Rainy season comes after the summer season. It is cloudy, wet and rainy. It rains heavily. It rains now and then.



Rainy Season

The weather is pleasant when it rains. People use umbrellas and raincoats when they go out of their houses.

People also wear gumboots. Sometimes there is a rainbow in the sky.

Winter Season : Winter season is cold and dry. It comes after the rainy season. People wear woollen clothes in winter season.

The sun is not bright. It is cold in winter. Snow falls on the Himalayas in winter season.

People love to drink hot drinks like hot milk, tea and coffee. People sit in the sun and in front of fire or heater.



Winter Season

KEY WORDS

Atmosphere	—	blanket of gases
Weather	—	changing condition of atmosphere
Season	—	condition of weather for a long time



Backpedal

- ▶ There is blanket of gases all around us. We call it **atmosphere**.
- ▶ The weather keeps changing hour to hour or day to day.
- ▶ The condition of weather for a long period of time is termed as **season**.
- ▶ There are three main seasons — summer, rainy and winter.



Objective Type Questions

Continuous Review

A. Tick (✓) the correct option:

- There is a blanket of _____ around the earth.

(i) gases	☐	(ii) water	☐	(iii) soil	☐
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- _____ keeps changing.

(i) Weather	☐	(ii) Time	☐	(iii) Season	☐
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- There are _____ seasons.

(i) one	☐	(ii) two	☐	(iii) three	☐
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B. Fill in the blanks with correct words:

1. The changing condition of atmosphere is called _____.
2. There are _____ seasons.
3. _____ season is hot.

Summer
three
weather

C. Write True or False:

1. There is atmosphere all around the earth.
2. Weather changes hour to hour.
3. In summer, we wear woollen clothes.

Subjective Type Questions

A. Answer in one word:

1. What is all around the earth?
2. What keeps on changing?
3. How many seasons are there?

B. Answer the following questions:

1. What is the atmosphere?

2. What is weather?

3. Write a note on summer season.

Think and Respond

- What if there is only one season?

Critical Thinking

Life skills and Values



Healthy Literacy

- How can we keep our body cool in summers?

FUNNY PROJECTS



- Name some foods of each season in your notebook.

Experiential Learning



TEST PAPER 1



A. Tick (✓) the correct option:

1. Things are of _____ kinds.

(i) one

☐

(ii) two

☐

(iii) three

☐

2. A seed is sown in the _____.

(i) sand

☐

(ii) cement

☐

(iii) soil

☐

3. Birds _____.

(i) crawl

☐

(ii) walk

☐

(iii) fly

☐

B. Fill in the blanks:

1. Living things _____.

2. Leaves are _____ of a plant.

3. Grains are called _____.

4. A house gives _____.

C. Write True or False:

1. Plants are not important for us.

2. All fruits have seeds in them.

3. Vegetables are grains.

4. Kidneys help in digestion.

D. Answer the following questions:

1. Compare the living and non-living things.

2. What are water plants?

3. How does a seed grow?

4. What are fruits?

5. Which animals live on trees?

6. What are the basic needs?

7. Name all the sense organs.



TEST PAPER

2



A. Tick (✓) the correct option:

1. Food gives us _____.
(i) power ☐ (ii) energy ☐ (iii) None of these ☐
2. A house can be _____.
(i) kutcha ☐ (ii) pucca ☐ (iii) Both of these ☐
3. The main source of water is _____.
(i) river ☐ (ii) pond ☐ (iii) rain ☐
4. We live on the _____.
(i) Earth ☐ (ii) Moon ☐ (iii) Sun ☐

B. Fill in the blanks:

1. A pucca house is _____.
2. Drink _____ of clean water.
3. Air occupies _____.
4. Winter season is _____.

C. Write True or False:

1. Do not do the exercise.
2. Never run down the stairs.
3. Air has mass.
4. Rainy season is pleasant.

D. Answer the following questions:

1. What is a balanced diet?
2. What is the meaning of cleanliness?
3. Why do we need clothes?
4. Why do accidents occur?
5. Why do we need water?
6. What do we breathe in?
7. What is the full moon?
8. What is weather?